The documentation and process conversion measures necessary to comply with this revision shall be completed by 6 November 1999.

INCH-POUND

MIL-PRF-19500/337G
6 August 1999
SUPERSEDING
MIL-S-19500/337F
4 October 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING
TYPES 1N4153, 1N4153-1, 1N4153UR-1, AND 1N4534
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, switching diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500 (see 6.3.2).

1.2 Physical dimensions. See figure 1 (similar to D0-35 and D0-34) and figure 2 (D0-213AA).

1.3 Maximum ratings.

$V_{(BR)}$	V_{RWM} 1/	I_o 2/	I_{FSM} t_p = 1 s	I_{FSM} t_p = 1 µs	T_{op} and T_{STG}	$Z_{\theta JX}$	$R_{\theta JL}$	$R_{\theta JEC}$
V dc	V (pk)	mA dc	A (pk)	A (pk)	°C	°C/W	°C/W	°C/W
75	50	150	0.25	2.0	-65 to +175	70	250	100

1/ This voltage may be exceeded for normal measuring intervals (< 1 minute) provided the current is limited to a maximum of 100 µA dc.

2/ Derate at 1.0 mA dc/°C above T_A = 25°C.

1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise indicated.

Limits	V_{F1} I_F = 100 µA dc	V_{F2} I_F = 250 µA dc	V_{F3} I_F = 1 mA dc	V_{F4} I_F = 2 mA dc	V_{F5} I_F = 10 mA dc	V_{F6} I_F = 20 mA dc
Min	0.490 V dc	0.530 V dc	0.590 V dc	0.620 V dc	0.700 V dc	0.740 V dc
Max	0.550 V dc	0.590 V dc	0.670 V dc	0.700 V dc	0.810 V dc	0.880 V dc

Limit	I_{R1} V_R = 50 V dc	I_{R2} V_R = 50 V dc T_A = 150°C	C V_R = 0 V dc f = 1 MHz	t_{rr} I_F = I_R = 10 mA dc R_L = 100 ohms, I_{RR} = 1.0 mA dc
Min				
Max	50 nA dc	50 µA dc	2.0 pF	4 ns

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 (similar to D0-35 and D0-34) and figure 2 (D0-213AA) herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 1N4153-1 construction. This device shall be constructed utilizing high temperature metallurgically bonded double plug construction in accordance with the requirements of category I,II, or III (see MIL-PRF-195OO).

3.3.3 JANS construction. JANS construction shall be metallurgically bonded, double plug, non-cavity thermally matched package. JANS metallurgical bonds are limited to category I only in accordance with MIL-PRF-19500.

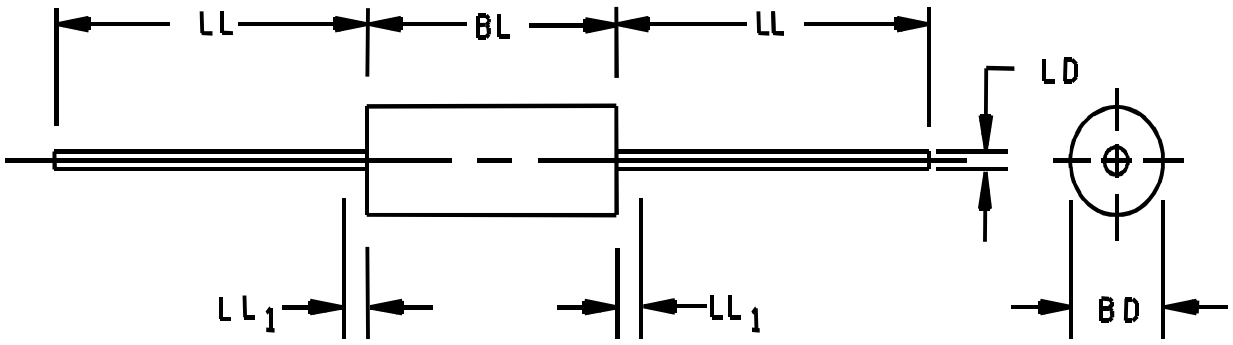
3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

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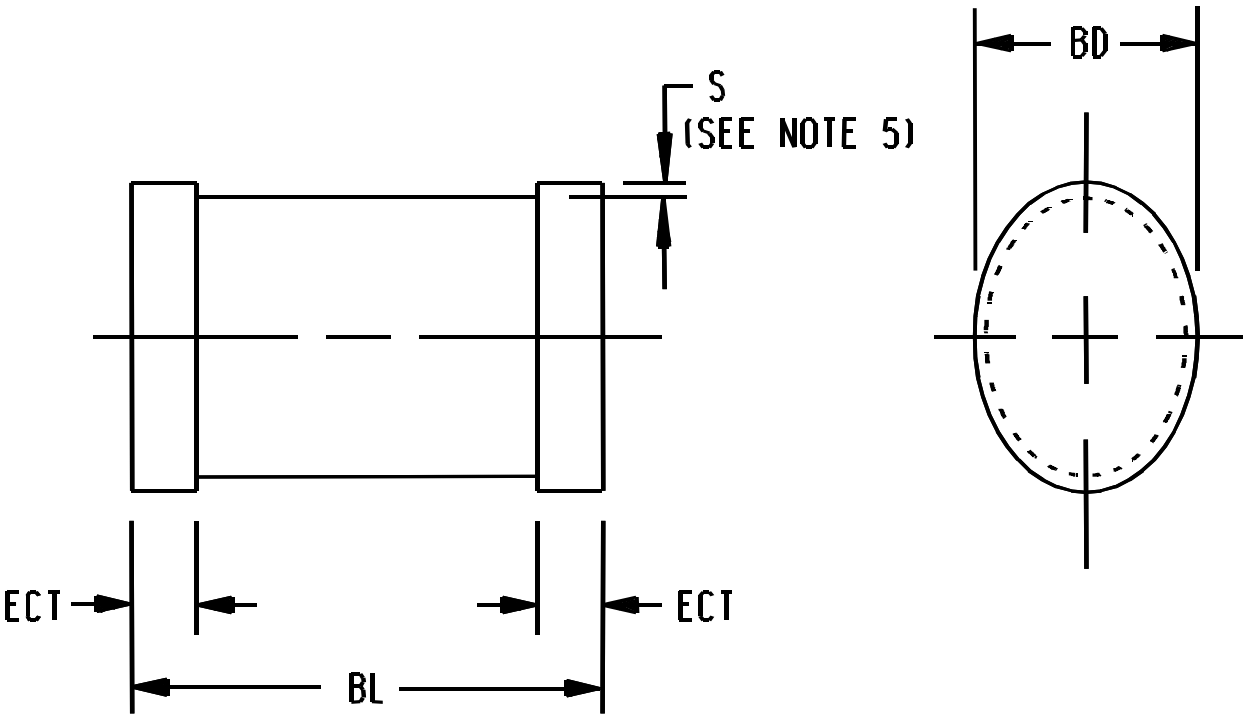


Types	Dimensions					Notes
	Ltr	Inches		Millimeters		
		Min	Max	Min	Max	
1N4153 1N4153-1	BL	0.140	0.180	3.56	4.57	3
	BD	0.056	0.075	1.42	1.91	3
	LL	1.000	1.500	25.40	38.10	
	LL_1		0.050		1.27	4
	LD	0.018	0.022	0.46	0.56	
1N4534	BL	0.080	0.120	2.03	3.05	3
	BD	0.050	0.075	1.27	1.91	3
	LL	1.000	1.500	25.40	38.10	
	LL_1		0.050		1.27	4
	LD	0.018	0.022	0.46	0.56	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within BD and length BL. Heat slugs if any, shall be included within this cylinder but shall not be subject to the minimum limit of BD.
4. Lead diameter not controlled in zones LL_1 to allow for flash, lead finish build-up and minor irregularities other than heat slugs.

FIGURE 1. Physical dimensions for 1N4153, 1N4153-1 (D0-35) and 1N4534 (D0-34).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	0.063	0.067	1.60	1.70
ECT	0.016	0.022	0.41	0.55
BL	0.130	0.146	3.30	3.70
S	0.001 min		0.03 min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions for 1N4153UR-1 (D0-213AA).

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4.VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.3 Screening (JAN, JANTX, JANTXV and JANS levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement		
	JANS	JANTX and JANTXV levels	JAN Level
3a	Temperature cycling	Temperature cycling	Temperature cycling (per MIL-PRF-19500 JANTX level)
3c 1/	Thermal impedance (see 4.5.2)	Thermal impedance (see 4.5.2)	Thermal impedance (see 4.5.2)
9	I_{R1} and V_{F6}	Not applicable	Not applicable
11	I_{R1} ; V_{F6} ; $\Delta I_{R1} \leq$ 100 percent of initial value or 25 nA dc, whichever is greater; $\Delta V_{F6} \leq$ 25 mV dc change from initial value	I_{R1} and V_{F6}	
12	See 4.3.1	See 4.3.1, t = 48 hours (min)	
13 2/ 3/	Subgroups 2 and 3 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial value or 25 nA dc, whichever is greater; $\Delta V_{F6} \leq$ 25 mV dc change from initial value.	Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial value or25 nA dc, whichever is greater; $\Delta V_{F6} \leq$ 25 mV dc change from initial value.	

1/ Thermal impedance may be performed any time after temperature cycling is completed in screen 3.

2/ Thermal impedance does not need to be repeated in step 13 if previously done in screening.

3/ PDA = 5 percent.

4.3.1 Power burn-in conditions. Devices shall be operated under one of the following conditions:

T_A = Room ambient as defined in the general requirements of MIL-STD-750, 4.5.

Condition 1: I_O = 150 mA dc, V_R = 50 V (pk).

Condition 2: I_F = 175 mA dc.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. (End-point electrical measurements shall be in accordance with table I, subgroup 2 herein.)

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and paragraphs 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	4066	Not applicable.
B4	1037	I_O = 150 mA dc; V_R = 50 V (pk); T_A = 25 °C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles.
B5	1027	I_O = 150 mA dc; V_R = 50 V (pk); T_A = 125°C, t = 96 hrs.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	I_O = 150 mA dc, V_R = 50 V (pk), T_A = Room ambient as defined in the general requirements of MIL-STD-750, 4.5.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C2	2036	(Not applicable for UR suffix version devices.) Test condition A; weight = 4 pounds.; t = 15 s. Test condition E; weight = 16 oz.
C5	4081	$R_{\theta JL}$ at L = 0.375 ≤ 250°C/W, $R_{\theta JEC}$ at L = 0 ≤ 100°C/W.
C6	1026	I_O = 150 mA dc, V_R = 50 V (pk). T_A = Room ambient as defined in the general requirements of MIL-STD-750, 4.5.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions of pulse measurements shall be specified in paragraph 4.3.2.2 of MIL-STD-750.

4.5.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 and as follows:

a. I_M measurement current.....1 mA - 10 mA.

b. I_H forward heating current .....300 mA to 500 mA.

c. t_H heating time.....10 ms.

d. t_{MD} measurement delay time .....100 µs maximum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3101	See 4.5.2	$Z_{\theta JX}$		70	°C/W
Forward voltage	4011	I_F = 100 µA dc	V_{F1}	0.49	0.55	V dc
Forward voltage	4011	I_F = 250 µA dc	V_{F2}	0.53	0.59	V dc
Forward voltage	4011	I_F = 1 mA dc	V_{F3}	0.59	0.67	V dc
Forward voltage	4011	I_F = 2 mA dc	V_{F4}	0.62	0.70	V dc
Forward voltage	4011	I_F = 10 mA dc	V_{F5}	0.70	0.81	V dc
Forward voltage	4011	I_F = 20 mA dc	V_{F6}	0.74	0.88	V dc
Reverse current	4016	DC method; V_R = 50 V dc	I_{R1}		50	nA dc
Breakdown voltage	4021	I_R = 5 µA dc	V_{BR1}	75		V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Reverse current	4016	DC method, V_R = 50 V dc	I_{R2}		50	µA dc
Low temperature operation:		T_A = -55°C				
Breakdown voltage	4021	I_R = 5 µA dc	$V_{(BR)2}$	75		V dc
Subgroup 4						
Junction capacitance	4001	V_R = 0 V dc, f = 1 MHz V_{sig} = 50 mV (pk to pk), max	C		2.0	pF
Reverse recovery time	4031	Condition A; C ≥ 1 nF $I_F = I_R$ = 10 mA dc, $i_{R(REC)}$ = 1.0 mA R ≥ 1,000 Ω, R_L = 100 Ω	t_{rr}		4	ns
Subgroup 5						
Not applicable						

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 6						
Surge current	4066	I_F = 150 mA dc; T_A = 25°C, I_{FSM} = 2 A(pk), ten 1 µs surges, 1 surge/minute				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups A, B, and C delta electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward current	4011	I_F = 20 mA dc	ΔV_{F6}	≤ 25 mV dc change from initial value		
2.	Reverse current	4016	DC method, V_R = 50 V dc	ΔI_{R1}	≤ 100 percent of initial value or 25 nA dc, whichever is greater.		

1/ The delta electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroups 4 and 5, see table II herein, steps 1 and 2.

2/ The delta electrical measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroup 6, see table II herein, steps 1 and 2 (JANS).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements should be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.1.1).

b. The lead finish as specified (see 3.3.1).

c. Type designation and quality assurance level.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA – NA
DLA – CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG
Air Force - 19, 99

Preparing activity:
DLA – CC

(Project 5961 – 2132)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/337G	2. DOCUMENT DATE 990806

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING TYPES 1N4153, 1N4153-1, 1N4153UR-1, AND 1N4534 JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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